Masoud Hemmatpour

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PROFILE

Software and Systems Engineer specializing in **distributed systems, networking**, and **AI infrastructure**. 7+ years of experience designing high-performance data processing systems across research and industry environments. Experience collaborating with organizations such as **Cisco Systems**, **Nokia Bell Labs**, and **Dolphin Interconnect Solutions**. Strong background in **Linux systems**, **networking**, and **scalable computing platforms**.

PROFESSIONAL EXPERIENCE

- **Arctic University of Norway (UiT)** — Tromsø, Norway
  Research Engineer — *May. 2023 – Apr. 2025*
  - Designed and implemented an AI-driven framework that improved edge device energy efficiency by 4×, enabling more efficient distributed inference.
- **Simula Lab** — Oslo, Norway
  Research Engineer — *March. 2021 – Dec. 2023*
  - Developed in-network computing solutions to accelerate distributed data processing for applications including e-commerce analytics and smart grid monitoring.
- **Cisco Systems** — Paris, France
  Senior Software engineer — *Jan. 2019 – Jan. 2020*
  - Designed and implemented control and management plane solutions for cloud-native environments using C, Docker, Grafana, and Linux networking.

SELECTED PROJECTS

- **Horizon Europe MISO project**
  - Developed an intelligent framework that reduce energy consumption of end node up to 4 times.
  - Formulated work package ideas under our responsibility for project execution.
  - Participated in all steering, review, and technical meetings.
  - Delivered project deliverables and reports on time.
- **Linux Foundation hICN Project**
  - Designed and implemented models for software switch (i.e., VPP) to interface with network controllers, enabling remote telemetry and configuration capabilities.
  - Project demonstrated in Cisco Live.

ACHIEVEMENTS

- Founder of COM4AI, a community focused on AI and communication systems.
- Pioneered memory storage over modern PCIe interconnect in HPC infrastructure.
- Developed a holistic fall prediction and detection system as part of the OPLON project.
- Best PhD dissertation award at Politecnico di Torino.
- Distinguished provincial mathematical prize 2000.

TECHNICAL SKILLS

Programming Languages
C/C++
Python
Bash

Systems & Networking
Ethernet
InfiniBand
RoCE

Operating Systems
Ubuntu(Debian)
CentOS(Red Hat)
Windows

Data Storage
MySQL
Memcached
Redis

Distributed Systems
POSIX threads
OpenMP
MPI

Machine Learning
PyTorch
STM32CubeAI
In-network ML

DevOps/Automation
CI/CD Pipelines
GitHub Actions

Version Control
Git
Gerrit

Monitoring
YANG/sFlow
Grafana

Containerization
Docker
Kubernetes

PRACTICAL TRAINED COURSES

MCSA (Microsoft Certified System Administrator): a certification of the ability to configure or manage a windows server infrastructure or desktops and devices in an organization.

LPI (Linux Professional Institute) courses: vendor-independent professional certification for Linux system administrators and programmers.

CCNA (Cisco Certified Network Associate): a certification of the ability to install, configure, operate, and troubleshoot medium-size routed and switched network.

EDUCATION

- **Politecnico di Torino** — Torino, Italy
  PhD in computer engineering — *Nov. 2015 – Sept. 2019*
  - **Thesis**
    - High Performance Computing using InfiniBand based clusters
  - **GPA: Graduated with distinction, receiving excellent evaluation from the thesis jury.**
- **Politecnico di Torino** — Torino, Italy
  MS in communications and computer networks engineering — *Oct. 2012 – Oct. 2015*
  - **Thesis**
    - Read-copy update synchronization technique improvement by hardware message passing
      Research project at École Politechnique Fédérale de Lausanne (EPFL)
  - **GPA: ≈ 3.93/4**

LANGUAGE SKILLS

- Norwegian, Intermediate, **B1**
- Italian, Upper Intermediate, **B2**
- English, **Fluent**
- Persian, **Native**

REFERENCES

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